

Research ethics and consent

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Effect size and practical importance

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Research ethics and consent. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Research ethics and consent. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Proposal automatically copies student essays into a study because they were submitted for grading. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: This classroom policy exercise does not determine real legal or institutional requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled research ethics and consent evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of research ethics and consent. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Research ethics and consent

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

Consent covers this study only. It excludes public posting of originals and unrelated face-recognition research.

Research ethics and consent

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State the purpose

Explain the drawing task and planned use before participation.



2. Respect exclusions

Do not publish originals or transfer them to the unrelated project.



3. Keep the agreement

Record permitted uses and the approved withdrawal procedure.

CHANGE ONE THING

A new project requests the same drawings for an excluded purpose.

Explain the changed result using the matching lesson.

Identify which uses the drawing consent permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record permitted uses and the approved withdrawal procedure.

1. State the purpose
2. Keep the agreement
3. Respect exclusions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep the agreement Record permitted uses and the approved withdrawal procedure. Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Practice 2: explain the deciding evidence

Changed-case practice: A new project requests the same drawings for an excluded purpose. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Do not reuse them under the existing consent; seek the required new authorisation.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Do not reuse them under the existing consent; seek the required new authorisation. The new purpose falls outside the stated agreement.

Repair a missing building block

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the mechanism and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the mechanism and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the mechanism and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

